

Computer Vision Mini project Teacher Assessment-2

Name - Kunal Kothe

Roll - A4-71

```
import cv2
import matplotlib.pyplot as plt

def sift_keypoint_matching(img1_path, img2_path):
    # Load images in grayscale
    img1 = cv2.imread(img1_path, cv2.IMREAD_GRAYSCALE)
    img2 = cv2.imread(img2_path, cv2.IMREAD_GRAYSCALE)

    # Initialize SIFT detector
    sift = cv2.SIFT_create()

    # Detect keypoints and descriptors
    keypoints1, descriptors1 = sift.detectAndCompute(img1, None)
    keypoints2, descriptors2 = sift.detectAndCompute(img2, None)

    # Use BFMatcher to find best matches
    bf = cv2.BFMatcher(cv2.NORM_L2, crossCheck=True)
    matches = bf.match(descriptors1, descriptors2)

    # Sort matches by distance (lower distance is better)
    matches = sorted(matches, key=lambda x: x.distance)

    # Draw matches
    match_img = cv2.drawMatches(img1, keypoints1, img2, keypoints2, matches[:50], None, flags=cv2.DrawMatchesFlags_NOT_DRAW_SINGLE_POINTS)

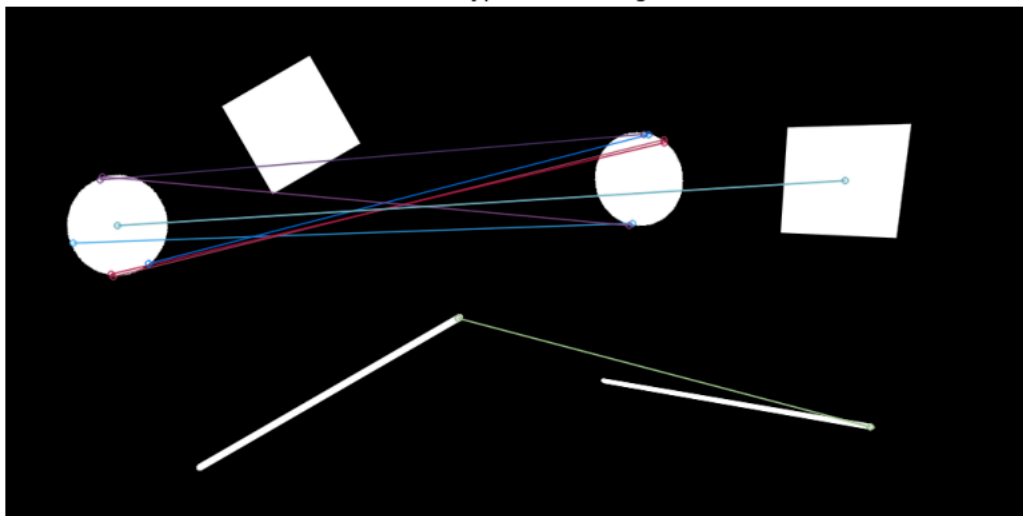
    # Show the result
    plt.figure(figsize=(10, 5))
    plt.imshow(match_img, cmap='gray')
    plt.title('SIFT Keypoint Matching')
    plt.axis('off')
    plt.show()

# File paths
img1_path = "sift_rotated.png"
img2_path = "sift_perspective.png" # Change as needed

# Run SIFT matching
sift_keypoint_matching(img1_path, img2_path)
```



SIFT Keypoint Matching



Start coding or [generate](#) with AI.